1.4.1 Conditioning of a linear system ¶

01.4.1 The condition number of a matrix

fit width

A question we will run into later in the course asks how accurate we can expect the solution of a linear system to be if the right-hand side of the system has error in it.

Formally, this can be stated as follows: We wish to solve $Ax = b$, where $A \in \mathbb{C}^{m \times m}$ but the right-hand side has been perturbed by a small vector so that it becomes $b + \delta\!b$.

> **Remark 1.4.1.1.** Notice how the δ touches the b. This is meant to convey that this is a symbol that represents a vector rather than the vector b that is multiplied by a scalar δ.

The question now is how a relative error in b is amplified into a relative error in the solution x.

This is summarized as follows:

$$\begin{array}{rcll} Ax & = & b & \quad \text{exact equation} \\ A(x + \delta\!x) & = & b + \delta\!b & \quad \text{perturbed equation} \end{array}$$

We would like to determine a formula, $\kappa(A, b, \delta\!b)$, that gives us a bound on how much a relative error in b is potentially amplified into a relative error in the solution x:

$$\frac{\|\delta x\|}{\|x\|} \leq \kappa(A, b, \delta b)\frac{\|\delta b\|}{\|b\|}.$$

We assume that A has an inverse since otherwise there may be no solution or there may be an infinite number of solutions. To find an expression for $\kappa(A, b, \delta b)$, we notice that

$$\begin{array}{rcl} Ax + A\delta x & = & b + \delta b \\ Ax & = & b \quad - \\ \hline A\delta x & = & \delta b \end{array}$$

and from this we deduce that

$$\begin{array}{rcl} Ax & = & b \\ \delta x & = & A^{-1}\delta b. \end{array}$$

If we now use a vector norm $\|\cdot\|$ and its induced matrix norm $\|\cdot\|$, then

$$\begin{array}{rcl} \|b\| & = & \|Ax\| \leq \|A\|\|x\| \\ \|\delta x\| & = & \|A^{-1}\delta b\| \leq \|A^{-1}\|\|\delta b\| \end{array}$$

since induced matrix norms are subordinate.

From this we conclude that

$$\frac{1}{\|x\|} \leq \|A\|\frac{1}{\|b\|}$$

and

$$\|\delta x\| \leq \|A^{-1}\|\|\delta b\|$$

so that

$$\frac{\|\delta x\|}{\|x\|} \leq \|A\|\|A^{-1}\|\frac{\|\delta b\|}{\|b\|}.$$

Thus, the desired expression $\kappa(A, b, \delta b)$ doesn't depend on anything but the matrix A:

$$\frac{\|\delta x\|}{\|x\|} \leq \underbrace{\|A\|\|A^{-1}\|}_{\kappa(A)}\frac{\|\delta b\|}{\|b\|}.$$

Definition 1.4.1.2. Condition number of a nonsingular matrix. The value $\kappa(A) = \|A\|\|A^{-1}\|$ is called the condition number of a nonsingular matrix A.

A question becomes whether this is a pessimistic result or whether there are examples of b and δb for which the relative error in b is amplified by exactly $\kappa(A)$. The answer is that, unfortunately, the bound is tight.

- There is an $\widehat{x}$ for which

$$\|A\| = \max_{\|x\|=1} \|Ax\| = \|A\widehat{x}\|,$$

namely the x for which the maximum is attained. This is the direction of maximal magnification. Pick $\hat{b} = A\widehat{x}$.

- There is an $\widehat{\delta b}$ for which

$$\|A^{-1}\| = \max_{\|x\|\neq 0} \frac{\|A^{-1}x\|}{\|x\|} = \frac{\|A^{-1}\widehat{\delta b}\|}{\|\widehat{\delta b}\|},$$

again, the x for which the maximum is attained.

It is when solving the perturbed system

$$A(x + \delta x) = \hat{b} + \widehat{\delta b}$$

that the maximal magnification by $\kappa(A)$ is observed.

Homework 1.4.1.1. Let $\|\cdot\|$ be a vector norm and corresponding induced matrix norm.

TRUE/FALSE: $\|I\| = 1$.

▼ Answer ▼ Solution

$$\|I\| = \max_{\|x\|=1} \|Ix\| = \max_{\|x\|=1} \|x\| = 1$$

TRUE

Homework 1.4.1.2. Let $\|\cdot\|$ be a vector norm and corresponding induced matrix norm, and A a nonsingular matrix.

TRUE/FALSE: $\kappa(A) = \|A\|\|A^{-1}\| \geq 1$.

▼ Answer ▼ Solution

$$
\begin{aligned}
&1\\
&\quad = \quad \langle \text{ last homework } \rangle\\
&\|I\|\\
&\quad = \quad \langle\ A \text{ is invertible } \rangle\\
&\|AA^{-1}\|\\
&\quad \leq \quad \langle\ \|\cdot\| \text{ is submultiplicative } \rangle\\
&\|A\|\|A^{-1}\|.
\end{aligned}
$$

TRUE

Remark 1.4.1.3. This last exercise shows that there will always be choices for b and $\delta\! b$ for which the relative error is at best directly translated into an equal relative error in the solution (if $\kappa(A) = 1$).